LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Design and preparation of PCB layout for a given circuit
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Design and preparation of PCB layout for a given circuit.
5. Specific Objectives	24.12.1.1 Trainee will be able to design and prepare a PCB layout for a given circuit.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	PCB Layout Design and Preparation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a PCB layout?
(b) List three critical factors affecting PCB trace width.
(c) Why is component placement important before routing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Circuit schematic analysis • Component footprint selection • Component placement strategy • Trace width and spacing rules • Via placement and routing	L & Ds	PPT, IB	15 min
(b)	• Importing schematic to layout tool • Placing components and orienting • Routing power and ground nets • Routing signal nets • Design rule check (DRC) verification	L & Ds	PPT, PDF	20 min
(c)	• Final layout review and cleanup • Generating Gerber files • Exporting drill files • Documentation of design notes • Preparation for fabrication	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between trace width and current carrying capacity.
(b) Describe the steps taken to ensure the layout meets design rules.
(c) How does proper component placement affect signal integrity?

12. Home Assignment and Follow Up	Prepare a brief report on the design constraints encountered during the layout and propose solutions for future designs, bridging to LP 02 soldering techniques.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of Soldering and de-soldering techniques adopted on PCBs and cables. Safety precautions to be observed while Soldering and De-soldering on PCBs and cables.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min